

# SchoolPath Data Engineering Intern Take-Home

## Context

SchoolPath helps families understand schools beyond the headline rating. We care a lot about two things that are notoriously hard to quantify: how well a school serves students with special needs, and what a school is actually like to attend, meaning its culture, climate, and safety. Test scores are easy to find. These things are not. That is the interesting part, and it is the part we want to see how you handle.

## The Task

Build a clean dataset of K-12 schools in Dallas, TX (public, charter, and/or private, your call, but justify it) and a simple dashboard that visualizes it. Aim for a defensible, well-scoped set of schools (roughly 40 to 80) rather than all of Dallas. Quality and clarity beat raw coverage.

Your dataset should go well beyond name, address, and rating. We want you to assemble attributes across two themes:

### 1. Special education relevance

What would a parent of a child with an IEP, autism, a learning disability, or a physical disability want to know? Find data that speaks to how a school actually serves these students, not just whether a program nominally exists.

### 2. School culture and safety

What is the lived experience of this school? Climate, discipline practices, safety, inclusion, teacher stability, parent and student sentiment. Some of the most useful signals here are unstructured or unofficial, and finding them is part of the challenge.

We are intentionally not giving you a list of sources. A big part of what we are evaluating is your ability to discover where useful data hides, and to judge what is trustworthy and what is not.

# **Deliverables**

## **1. The dataset**

CSV or Parquet, one row per school, with a documented schema, meaning a data dictionary listing each column, its type, its source, what it means, and its coverage (percent of rows populated).

## **2. A dashboard**

Anything that runs and renders: Streamlit, a notebook with charts, Looker Studio, Observable, a small web app. It should let a non-technical person explore the data and walk away with insight. Visualize comprehensively, showing the special-ed and culture/safety dimensions, not just a map of dots.

## **3. A write-up (1 to 2 pages, in your own words)**

This must be written by you, not by an AI. We want to hear how you think. We will be able to tell, and a generic AI-generated write-up will count against you. Cover:

- Where you got the data and why you trusted it
- The hardest cleaning or normalization problem you hit and how you solved it
- At least one creative or non-obvious data source you used, and what it added
- What is missing, what you would add with another week, and what data you deliberately chose not to use, and why
- Any caveats: bias, staleness, coverage gaps, joins you were not confident about

You can use AI tools freely to build the project. Just write the write-up yourself.

## **4. A reproducible repo**

We should be able to clone your repo and rebuild the entire dataset from scratch by running your scripts, with no manual steps we have to guess at, and no "and then I cleaned it by hand in Excel." Every transformation from raw source to final dataset should be in code and runnable. Include:

- A README that gets us from zero to dashboard
- All collection, scraping, and fetching scripts

- All cleaning and joining logic, in order
- Clear instructions to reproduce (dependencies, commands, any keys or config needed)

If we run it, we should end up with the same dataset you submitted.

## **Ground rules**

- Be scrappy. That is the whole point. We want to see how far your resourcefulness goes. Do not let a paywall, a missing API, an awkward export, a locked PDF, or a "this data is not really available" wall stop you. Finding a way through obstacles is exactly the skill we are testing. We will not penalize aggressive or unconventional data gathering, and we would rather see you push limits than play it safe.
- Time budget: roughly 8 to 10 focused hours. We are not looking for polish on everything. We are looking for good judgment about where you spent your time. A smaller dataset with sharp insight and honest caveats beats a huge messy one.
- Use any language, stack, or tools you like. Tell us what you used.

## **How we evaluate**

We care about: research resourcefulness, how you reason about which data is worth having, data-cleaning rigor, the entity-matching and joining problem (matching the same school across sources is harder than it looks), reproducibility of your pipeline, and whether your dashboard actually communicates something. Above all, we want to see how you think about messy, hard-to-find data, and how scrappy you are willing to get to find it.